Mengjie Zhang

PhD Candidate in Biomechanics and Ergonomics · The University of Auckland

mengjie.zhang@auckland.ac.nz • mzha967@aucklanduni.ac.nz • ORCID 0009-0006-8016-8904
• mengjie-zhang.github.io • Auckland, New Zealand

Scan for the academic profile

2	9	3	4
First-author journal articles	Conference outputs, 2021–2026	Oral presentations in 2026, 2 at the World Congress of Biomechanics	Assessment modules in the online video-based system

Education

2023 – 2027 (expected)
PhD, Biomechanics and Ergonomics
The University of Auckland, New Zealand

2020 – 2023
Master's degree, Sport Human Science
Beijing Sport University, China

2016 – 2020
Bachelor's degree, Sport Human Science
Beijing Sport University, China

Scholarship

CSC doctoral scholarship (China Scholarship Council), Oct 2023 – Oct 2027, 48 months

Skills

MOTION CAPTURE
Vicon · Motion Analysis (Cortex) · OpenCap · MediaPipe

MODELLING
OpenSim · Visual3D · Inverse dynamics · Static optimisation · SQP

MACHINE LEARNING
PyTorch · LoRA fine-tuning · LOSO cross-validation

ERGONOMICS
RULA · REBA · Revised NIOSH · 3DSSPP

LABORATORY TESTING
Kistler force plates · Cosmed K4b2 · Isokinetics · sEMG

DATA AND SOFTWARE
Python · MATLAB · SPSS · Streamlit · ICC, RMSE, R^2

Languages

Chinese (native) · English (professional)

Research profile

Developing and validating computational methods, grounded in biomechanics research, for human movement analysis and ergonomic design. Building on training in sport science and sports biomechanics, my research spans digital human modelling, markerless motion capture, occupational biomechanics and data-driven musculoskeletal modelling.

As a research assistant, I am the lead developer of a markerless system for human movement analysis from ordinary single-camera video, supporting gait analysis, manual-handling risk assessment with lumbar loading estimation, and preliminary screening in rehabilitation assessment.

Research overview

DESIGN PHASE

Systematic review — Posture prediction models in DHM → **Hybrid model** — ≈85 ms per posture

EVALUATION PHASE

Markerless validity — OpenCap and MediaPipe vs Vicon → **Online system** — RULA/REBA, NIOSH, static and dynamic biomechanics → **Evaluation** — Reference methods and expert end users

Extension · single-camera musculoskeletal analysis: kinematics, deep-learning ground reaction forces, inverse dynamics and muscle activation

Journal articles

[1] **Zhang, M.**, Nieuwenhuys, A., Boocock, M.*, & Zhang, Y.* (2027). **Concurrent validity of markerless motion capture for occupational lifting: Kinematics and ergonomic risk assessment.** *Applied Ergonomics*, 138, 104866. doi.org/10.1016/j.apergo.2026.104866

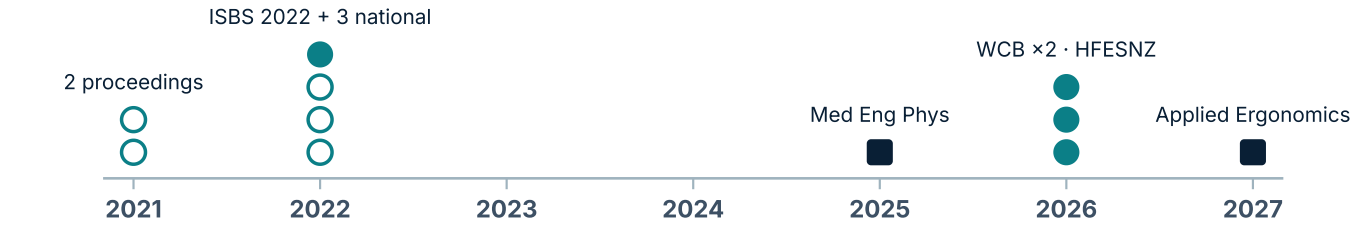
First author **Open access** Errors increased by 15–30% from symmetrical to asymmetric multi-planar tasks. Both systems were reliable for symmetrical lifting, including high REBA agreement, but validity declined for tasks involving trunk rotation and walking.

[2] **Zhang, M.**, Nieuwenhuys, A., & Zhang, Y.* (2025). **Posture prediction models in digital human modeling for ergonomic design: A systematic review.** *Medical Engineering & Physics*, 143, 104391. doi.org/10.1016/j.medengphy.2025.104391

First author **Open access** Following PRISMA guidelines, 24 studies were included from a search across nine academic databases, categorized into data-driven models (n = 12) and optimization-based models (n = 12).

* Corresponding author.

Research outputs, 2021–2027



Applied Ergonomics volume 138 (2027), published online August 2026.

Manuscripts

[1] **Zhang, M.**, Boocock, M., Nieuwenhuys, A., & Zhang, Y. **A hybrid two-phase framework for posture prediction of manual material handling tasks.** **Submitted to Ergonomics**

[2] **Zhang, M.**, Boocock, M., Nieuwenhuys, A., & Zhang, Y. **Development of an online video-based ergonomic assessment system.** **Manuscript drafted, being refined for submission**

Conference presentations and proceedings

[1] **Zhang, M.**, Boocock, M., Nieuwenhuys, A., & Zhang, Y. (2026). **From clipboard to camera: an online video-based ergonomic assessment system for New Zealand workplaces.** HFESNZ 40th Conference, Wellington, New Zealand. **Oral**

[2] **Zhang, M.**, Boocock, M., Nieuwenhuys, A., & Zhang, Y. (2026). **A Hybrid Two-Phase Framework for Posture Prediction of Manual Material Handling Tasks.** 10th World Congress of Biomechanics (WCB 2026), Vancouver, Canada. **Oral**

[3] **Zhang, M.**, Nieuwenhuys, A., Boocock, M., & Zhang, Y. (2026). **Concurrent Validity of Markerless Motion Capture for Assessing Manual Lifting Kinematics.** 10th World Congress of Biomechanics (WCB 2026), Vancouver, Canada. **Oral**

[4] **Zhang, M.**, & Li, H. (2022). **The effect of running economy in male runners wearing 3 types of footwear.** *Proceedings of the 40th Conference of the International Society of Biomechanics in Sports*, 40(1), Article 193.

[5] **Zhang, M.**, et al. (2022). **The impact of four racing shoes on the running economy of runners.** *Proceedings of the 22nd National Sports Biomechanics Academic Conference* (in Chinese).

[6] **Zhang, M.**, et al. (2022). **Differences in running economy and energy consumption in different marathon running shoes.** *Proceedings of the Seventh International Symposium on Sports and Health* (in Chinese).

[7] **Zhang, M.**, et al. (2022). **Effects of Different Types of Running Shoes on Running Economy—Visual Analysis.** The 2nd Human Sports Science Conference (in Chinese). **Oral**

[8] **Zhang, M.**, et al. (2021). **Research progress on the influence of strength training on the running economy of middle-distance runners.** *Medical Biomechanics*, 36(S8), 420 (in Chinese).

[9] Chen, X., **Zhang, M.**, & Li, H. (2021). *Proceedings of the 21st National Sports Biomechanics Academic Exchange Conference* (in Chinese).

Research experience

Research Assistant, video-based motion analysis system

Jun 2026 – present

The University of Auckland, New Zealand · PI: Associate Professor Yanxin Zhang

- Lead developer of a markerless system for human movement analysis from ordinary single-camera video, requiring no markers, force plates or laboratory set-up.
- Adapted the markerless pose model, including added neck degrees of freedom, and integrated video-derived kinematics with deep-learning ground reaction force prediction, inverse dynamics, muscle-activation estimation and a web interface.
- Fine-tuning a pretrained model for occupational lifting (low-rank adaptation with external-load conditioning) reduced vertical ground reaction force error by 46% relative to the zero-shot pretrained baseline (18.3% to 9.8% body weight RMSE) in 19-fold leave-one-subject-out cross-validation across 182 lifting trials (development-stage results).

Doctoral research

Online video-based ergonomic assessment system

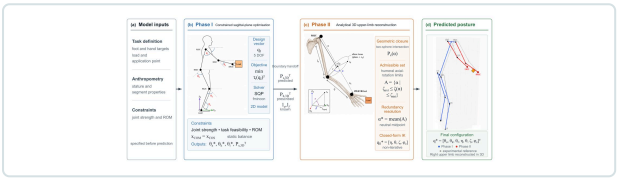
2023 – present

- Analyses workplace tasks from a single standard video using markerless pose estimation, removing the need for markers, wearable sensors, force plates or a laboratory. Four modules run from the same video: RULA and REBA postural scoring; NIOSH Recommended Weight Limit and Lifting Index; static joint moments and L5/S1 spinal compression; and dynamic biomechanics across full movement cycles.
- REBA and RULA risk categories agreed with the laboratory reference in 93.9% and 91.7% of 132 trials, the NIOSH Lifting Index showed a mean absolute error of 0.13, and against 3DSSPP, the L5/S1 moment and anteroposterior shear showed systematic offsets (+8.7% and −7.2%), whereas compression showed a small mean difference (+1.8%) but higher variability between trials. Four experts rated the system useful and easy to use.

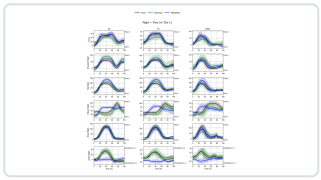
Hybrid two-phase posture prediction

WCB 2026

- Combines sagittal-plane whole-body optimisation with analytical 3D upper-limb reconstruction. Posture predictions were compared with marker-based motion capture in twelve participants; joint-angle RMSE ranged from 4.0° to 37.5°, with a total computation time of approximately 85 ms per posture.



Hybrid two-phase posture prediction framework.



Joint-angle waveforms: Vicon, OpenCap and MediaPipe.



Online system: live capture, highest-risk posture and PDF report.

Skills across projects

	Systematic review	Hybrid posture prediction	Markerless validity	Online assessment system	Single-camera MSK analysis	Footwear research
Marker-based motion capture		●	●	●	●	●
Markerless motion capture			●	●	●	
Force platforms and exercise physiology					●	●
Inverse dynamics and musculoskeletal modelling				●	●	●
Optimisation and digital human modelling	●	●				
Deep learning					●	
Ergonomic risk assessment (RULA, REBA, NIOSH)			●	●		
Validation statistics and evidence synthesis	●	●	●	●	●	●
Software development		●		●	●	

Earlier research and professional experience

Core Member, Footwear Performance Testing

Mar – Oct 2022

Beijing Sport University, China

- Responsible for testing the running economy of a racing-shoe series; carried out random crossover experiments on 15 male running enthusiasts, conducting the maximal oxygen uptake test, the running economy test and the biomechanical test of the lower limbs sequentially.

Intern, Sports Department

2021

Tsinghua University, China

- Assisted the supervisors to conduct physical fitness tests for physical education courses, including the Functional Movement Screen and body composition tests.

Project Lead, Master's Research

Nov 2021 – Jun 2022

Beijing Sport University, China

- Explored the effects of racing running shoes with different bending stiffness on energy metabolism and lower-extremity joint angles, angular velocities, moments, power and work using one-way repeated-measures analysis of variance, and the relationship between flexion stiffness, energy metabolism and mechanical work of the lower limbs using Spearman correlation.

Intern, Physical Fitness Testing for People with Disabilities

2021

Research project, Wuhan, China

- Responsible for physical fitness tests for people with disabilities, including height, weight, blood pressure, lung capacity, strength and speed.

Teaching and mentoring

- **Research mentoring** (2024 – present): postgraduate students in motion capture, data processing and analysis, and academic writing.
- **Teaching assistance** (2024 – present): tutorials and coursework marking.

Volunteering

Beijing Marathon (2017) · Beijing Tongzhou Marathon (2021) · Beijing Sport University COVID-19 vaccination programme (2021)